

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 1.5 RESPIRATION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit on Respiration. Return to the sealed thermos of uji that Achieng, a Form 4 student in Kisumu, opened at school. Use the evidence you collected in your investigations, your Driving Question Board and your model to answer each section in the space provided below it. Write in full sentences, use correct science words, and name the evidence that supports each idea. Where a question asks for an equation, write it clearly. Take your time, and remember to link every answer back to the thermos of uji.

Part 1: What is happening inside the sealed thermos?

Achieng poured fermented uji from a sealed thermos flask. She saw tiny bubbles and noticed that the uji smelled more sour than it did two hours earlier. No heat and no new ingredients were added. A jar of fresh uji wa kawaida (not fermented) did not change. (a) Explain what is producing the bubbles and the sour taste. (b) Use evidence from the yeast and lime water investigation (Flask A and Flask B) to support your explanation. (c) Explain why the process can continue inside a sealed flask.

(a) Living microorganisms already present in the fermented uji, mainly yeast and also bacteria, are carrying out cellular respiration. They break down sugar (glucose) from the maize porridge to release energy. Without oxygen they carry out anaerobic respiration (fermentation). The bubbles are carbon dioxide gas. The sour taste comes from acidic products that build up and lower the pH of the uji. The fresh uji wa kawaida did not bubble or change in the same way, which suggests it does not contain the active fermenting microorganisms that the fermented uji has. (b) In the investigation, Flask A (yeast + glucose solution) produced bubbles and turned the lime water milky within 10-20 minutes, showing that carbon dioxide was produced. Flask B (glucose only, no yeast) produced no bubbles and the lime water stayed clear. So the yeast, not the glucose alone, made the gas. Flask A also had a faint yeasty/ethanol smell and Flask B did not. (c) Anaerobic respiration does not need oxygen, so yeast can keep releasing energy from glucose even when the flask is sealed and no fresh air enters. Word equation: glucose → ethanol + carbon dioxide (+ a small amount of energy).

Part 2: Aerobic and anaerobic respiration compared

Amina, a sprinter in Nairobi, gets energy from the ugali she eats. (a) Write the word equation for aerobic respiration and state where in the cell it

(a) Glucose + oxygen → carbon dioxide + water + energy (ATP). It mainly happens in the mitochondria of cells. The starch in ugali is digested to glucose, which cells use as fuel. (b) Aerobic respiration needs a continuous supply of oxygen. The thermos is sealed so no fresh oxygen enters, yet bubbles still form. So another process, anaerobic respiration, which works without oxygen, must be occurring in the yeast. (c) In muscle cells with too little oxygen, glucose is broken down to lactic acid: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$ (glucose → lactic acid + a small amount of energy). The lactic acid causes the burning feeling in Amina's muscles. (d) Aerobic

mainly happens. (b) Explain why aerobic respiration alone cannot explain the bubbles in Achieng's sealed thermos. (c) Write the word equation or chemical equation for what happens in Amina's muscle cells when oxygen supply is not enough, and say what Amina feels. (d) Compare the amount of ATP released by the two types of respiration from the same amount of glucose, and say which releases more.	respiration releases much more ATP than anaerobic respiration from the same amount of glucose. The unit did not give exact ATP numbers, so the comparison is: aerobic = much more, anaerobic = only a small amount. Both anaerobic pathways (lactic acid in animals, alcoholic fermentation in yeast) release far less ATP and do not need oxygen.
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Part 3: Evidence for sourness, substrates and RQ

(a) In the universal indicator test, fresh uji, fermented uji from the sealed thermos, mala and a distilled water control were compared. Describe what the results showed for the uji samples and the distilled water, and explain what they tell you about the cause of sourness. Say what the mala result suggests. (b) Name the three main respiratory substrates, give a Kenyan food example of each, and state the RQ values you learned. (c) Write the formula for RQ. (d) The RQ is worked out for cells that are using oxygen. Explain what happens to the RQ value if anaerobic respiration is also contributing CO₂ in addition to aerobic respiration, and what that tells you about the type of respiration.	(a) Fermented uji had a pH of 3-4 (orange-red with universal indicator). Fresh uji was about pH 5.5-6.5 (yellow-green) and distilled water was pH 7 (green). So fermentation made the uji more acidic. Acidic by-products of anaerobic respiration lower the pH, and the tongue detects this as sourness. Mala, which is made by lactic acid bacteria, was also sour, which suggests acid from bacterial fermentation can cause sourness in the same way. (b) Carbohydrates (RQ = 1.0), for example ugali; fats (RQ about 0.7), for example groundnut oil or avocado; proteins (RQ about 0.8), for example nyama choma or beans. (c) $RQ = \frac{\text{volume of CO}_2 \text{ produced}}{\text{volume of O}_2 \text{ consumed}}$. (d) If anaerobic respiration adds CO₂ without any extra O₂ being used, the ratio becomes larger, so the RQ goes above 1.0. An RQ above 1.0 shows that anaerobic respiration is contributing to energy release. In the completely sealed thermos, once oxygen has run out no O₂ is consumed, so an RQ cannot be calculated there; the yeast are simply fermenting glucose, a carbohydrate, and the bubbles are the CO₂ they release.
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Part 4: Why does uji ferment at different rates?

Achieng's uji was kept in a thermos. Imagine the same uji in Mombasa (hot, humid coast) and in Kericho (cool highlands). (a) Predict where it would ferment faster and explain why, using what you found in the yeast investigation. (b) Name two other factors besides temperature that affect the rate of respiration and describe their effect with evidence from the class results. (c) Explain why a farmer in Kenya would keep stored maize cool and dry.

(a) It would ferment faster in Mombasa. Higher temperature increases enzyme activity in yeast, so glucose is broken down faster, producing CO₂ and acids more quickly. In our investigation, yeast-glucose solution at about 40 °C turned lime water milky in about 4 minutes, while at about 5 °C lime water barely changed in 15 minutes. In cool Kericho the enzymes work more slowly, so the uji stays fresh longer. (Too high a temperature would denature the enzymes.) (b) Substrate concentration: a higher glucose concentration (10 g/50 mL) gave faster, more sustained bubbling than a low concentration (2 g/50 mL), because there is more fuel. Surface area and metabolic activity: germinating maize seeds turned lime water milky significantly faster than the same number of dry seeds, showing that metabolically active cells respire faster. (c) Cool, dry conditions slow down respiration by microorganisms and the grain itself, so the grain is not spoiled and keeps longer.

Part 5: Using respiration to solve community problems

Answer the driving question: How do living organisms, from the yeast in our uji to the cells in our own muscles, release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities? (a) Write a three-sentence answer. (b) Give TWO Kenyan examples of anaerobic respiration used in the economy, naming the organism or product involved and the by-product that is useful. (c) Explain why brewing alcoholic drinks is not a responsible application of what you learned.

(a) All living organisms release energy from food (substrates such as glucose from ugali) through cellular respiration: aerobic respiration uses oxygen in the mitochondria to make carbon dioxide, water and a lot of ATP, while anaerobic respiration, such as yeast fermentation in the uji thermos or lactic acid production in muscles, works without oxygen and releases less ATP. The rate of respiration depends on factors like temperature and substrate concentration, and the products can be recognised by tests such as lime water for CO₂ and universal indicator for acidity. We can harness these processes to make useful products and solve problems, such as making uji, mala and yoghurt, leavening mandazi, making silage and producing biogas. (b) Example 1: Mala/yoghurt production, where lactic acid bacteria produce lactic acid, which gives sourness and preserves the milk. Example 2: Mandazi dough, where yeast produces CO₂ that makes the dough rise. Other acceptable examples: Rift Valley biogas, which gives fuel from anaerobic breakdown of dairy waste; silage, where lactic acid preserves animal feed. (c) Brewing alcoholic substances is excluded from responsible use of this knowledge because it can harm health and the community, so we use fermentation for safe food, feed and fuel.

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Explaining the anchoring phenomenon	Clearly explains that yeast (and bacteria) respire anaerobically in the sealed thermos, linking bubbles to CO ₂ and sourness to acid products, and uses both the Flask A/Flask B and pH evidence accurately.	Explains that microorganisms cause the bubbles and sourness but uses evidence from only one investigation or with small gaps.	States that something changes in the uji but does not link it to respiration or to evidence.
Aerobic and anaerobic respiration equations and comparison	Writes correct word or chemical equations for aerobic respiration, alcoholic fermentation and lactic acid fermentation, and correctly compares oxygen need and ATP released.	Writes most equations correctly and makes a basic comparison with a minor error or omission.	Equations are missing or incorrect and the comparison is unclear.
Substrates, RQ and rate factors	Correctly names substrates with Kenyan examples and RQ values, uses the RQ formula, interprets RQ above 1.0, and explains temperature and glucose concentration effects using class data.	Gives most substrate, RQ and factor information correctly, with limited use of data.	Gives few correct facts about substrates, RQ or rate factors and does not use data.
Application to Kenyan economy and community	Gives two accurate Kenyan applications naming organism, product and benefit, and explains the responsible-use caution on brewing.	Gives two applications with some missing detail, or a limited explanation of the caution.	Gives one or no application or the applications are incorrect.
Scientific communication and use of evidence	Writes in clear, organised sentences, uses scientific terms correctly, and links every part back to the uji thermos.	Writing is mostly clear with mostly correct terms and some links to the thermos.	Writing is unclear, terms are misused and links to the thermos are missing.

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